

Table 1: Permutation test for constant weights across true weighting shapes. The null hypothesis is that the latency weight is constant across lags ($\alpha(0) = \dots = \alpha(\tau)$). Working model: knot-search restricted cubic spline selected by AIC; for each replicate the observed AIC is compared with the null distribution from $B = 300$ permutations of the lag columns, and H_0 is rejected when the observed AIC falls below the lower $\alpha = 0.05$ percentile of that distribution. The rejection rate is taken over 300 simulated data sets ($n = 5000$, $\gamma = 0.1$, $\lambda_0 = 0.5$, true latency 16); for the constant shapes it estimates the type-I error and for the lag-varying shapes the power.

True weight shape	H_0 status	Replicates	Rejection rate
Unimodal	False	300	1.000
Damped exponential	False	300	1.000
Bimodal 1	False	300	1.000
Bimodal 2	False	300	1.000
Null effects	True	300	0.010
Constant positive effects	True	300	0.010